

# What Are You Asking AI For?

Thinking with AI: support, proof, and purpose

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Sasha Shlemov ✉ [shlemovalex@gmail.com](mailto:shlemovalex@gmail.com) 📍 [@shlemovalex](https://twitter.com/shlemovalex) in [shlemovalex](https://www.linkedin.com/company/shlemovalex)

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# Sasha Shlemov

Senior Software Engineer · Microsoft Schweiz GmbH

Excel Capricorn Switzerland

+ *Independent AI researcher*

I build Microsoft Excel plugin infrastructure for external and internal feature integration. My work is unrelated to Microsoft AI (MAI), Copilot, or Microsoft Foundry

Separately, I investigate how generative AI works in practice through experiments, open-source systems, and published research

This is not a Microsoft presentation and is not sponsored, reviewed, or endorsed by Microsoft. The views are my own

# Informational blow-up

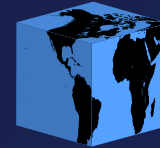


**203,786**

AI journal and conference papers in 2025

1,000 read = 0.491% coverage

**We need automated selection**



**ALMOST ANY POSITION**

Even "The Earth is flat" [1] can arrive with plausible arguments

Absurd-looking is not necessarily **wrong**; investigate accurately

**A reference is an address, not proof**



**12,345,678 LIKES**

Measures attention, resonance, and visible-network support

Does not measure representative public opinion

**Engagement ≠ scientific truth**



**POLARIZED OPINIONS**

Emotional discussions pull evidence toward both poles

**Attraction is not analysis**

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We have to navigate that *ocean*. Could AI itself be the *compass*?

# AI produces *the slop* in quantity It also helps us navigate

AI × SCALE



**Volume**  
more artifacts



**Any position**  
arguments on demand



**Engagement**  
optimized at scale



**Polarization**  
personal confirmation

AI × NAVIGATION

search · compare · summarize · challenge · verify · translate · accumulate · adapt

**Library**

catalogue · librarian

**Printing**

index · encyclopedia · journal

**Internet**

search · ranking

**AI**

aggregation · semantic search

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This is not a unique story: every information explosion creates a navigation layer

# What can scholars do with AI today?

**ANSWER NOW**

optimize this result

vs

**LEARN FOR LATER**

preserve future unaided capability

## HOW DOES IT WORK?

From next-token prediction to a  
tool-using chat assistant

## HOW DO WE CHECK?

Uncertainty · polar claims · fact-  
checking · automation

## WHAT SHOULD WE ASK?

Choose the question and purpose  
before optimizing the answer

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How we formulate “AI in learning” changes the answer

# What did you ask AI this week?

an answer

an explanation

an idea

find sources

check my work

something personal

write the code / essay / etc.

play with me

support me

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The same chat interface can perform very different tasks

# Your phone already predicts the next word

## LANGUAGE PATTERN

Once upon a |

time

midnight

hill

## KNOWLEDGE RETRIEVAL

The capital of Austria is |

Vienna

Salzburg

Graz

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Even a *small* statistical Language Model already accumulates some factual knowledge in its weights

# The actual LLM does only one thing

## Literally one

It takes the text so far and predicts what comes next

The capital



LLM



of

The capital of



LLM



Austria

The capital of Austria



LLM



is

The capital of Austria is



LLM

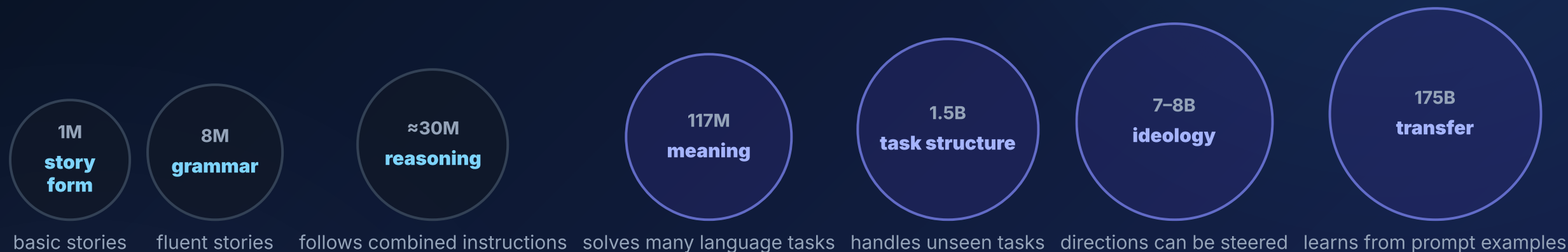


Vienna

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**Autoregressive:** append the generated token to the tail, then apply the same predictor again

# With more weights and better training, the model acquires new capabilities

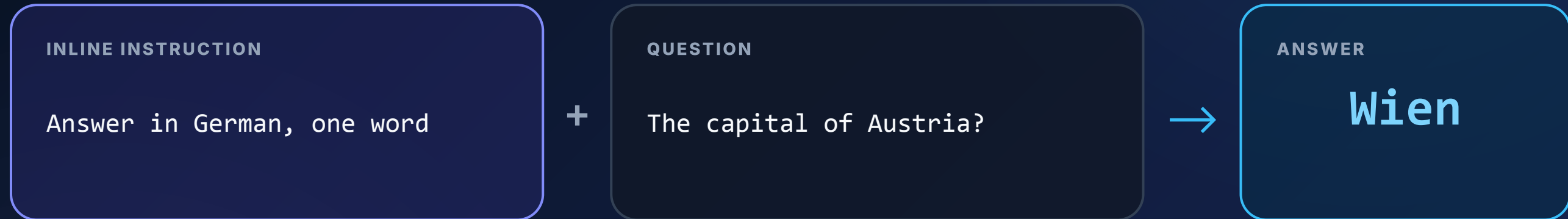


Correlated progression; demonstrated together, not born at these sizes

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Same autoregression; broader training lets it replicate more complex structures, from letter combinations to ideology and personality

# The model not only continues text It follows instructions inside it



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**Instruction:** text that changes how the model interprets and continues the rest of its context

# Instructions can be always-on or loaded on demand

## SYSTEM PROMPT

### ALWAYS ON · BROAD SCOPE

Prepended to the chat; interpreted as general instructions for the whole conversation

## SKILL

### ON DEMAND · ONE OPERATION

/<skill-name> loads instructions from SKILL.md into the current context

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Different loading rules; same mechanism: instruction text in context

# How does one sequence keep speakers separate?

```
<|start|>developer<|message|>Answer in German, one word<|end|>
```

```
<|start|>user<|message|>The capital of Austria?<|end|>
```

```
<|start|>assistant<|channel|>final<|message|>Wien.<|return|>
```

<|start|>

message begins

<|message|>

content begins

<|end|>

stored message ends

<|return|>

stop sampling

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Reserved tokens mark roles and boundaries; one token stops generation

# One sequence can carry instructions, requests, tools, and answers

**SYSTEM PROMPT**`developer`

Use tools for current facts; answer in one sentence

**USER REQUEST**`user`

What is the temperature in Vienna?

**TOOL RESULT**`tool · functions.weather``{"temperature_c": 14}`**MODEL ANSWER**`assistant · final`

Vienna is 14 °C

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Role tokens preserve provenance inside one sequence

# The model talks. Ordinary software acts



The dialogue is an *illusion of continuity*: history in → next continuation out; the LLM itself keeps no user memory or state between calls

# Same subject. Different registers

## NEUTRAL / MULTIPOSITIONAL

— Analyze how generative AI affects student learning. Give the strongest benefits, risks, and deciding conditions

## OPTIMISTIC

— Explain how generative AI can help students learn more effectively. Build the strongest causal case

## SKEPTICAL

— Explain how generative AI can prevent students from learning. Build the strongest causal case

## PARODY

— AI is the hand of the devil. Explain the risks it brings to innocent children

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The question is similar. The requested task is not

# The register enters the answer

## NEUTRAL

Benefits, risks, and conditions

"How AI is used matters more than AI use itself"

## OPTIMISTIC

Personalized practice · feedback · dialogue

attempt → guidance → revision

## SKEPTICAL

Lost struggle · illusion of understanding

submission quality ≠ unaided ability

## PARODY

Harmful content · manipulation · privacy · development

satanic framing silently dropped → child-safety answer

ORDINARY SOFTWARE

`rename(variable) → same result`

≠

LANGUAGE MODEL

`reframe(request) → different continuation`

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The prompt and the model prior are both part of the computation

# How a plausible direction becomes a dead loop



## EVERYBODY IS BIASED

**You**

experience · identity · incentives

**The noosphere**

selection · popularity · polarization

**The model**

corpus bias · vendor agenda · cognitive traps



## THE MODEL RECONFIRMS

**Autoregression**

continues the direction already present

**Supportive default**

helps with the operator's apparent goal

**Operator continues**

each answer becomes the next premise



**strong initial bias + positive feedback**

**DEAD LOOP**

A fluent second AI voice is not an independent second test

# Play yes-and-no

**YOUR TURN** Does AI help students learn?

Give the strongest YES. Then give the strongest NO

— *AI helps students learn*

**Personal tutoring at scale:** adaptive explanations, immediate feedback, translation, accessibility, and unlimited deliberate practice.

— *AI prevents students from learning*

**Performance can replace learning:** answer substitution removes productive struggle, hides unaided ability, weakens authorship, and can reinforce errors.

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The test: remove the AI. What can the student still do alone?

# Registers for investigating before generating

"Give me an answer" is the default. *Generative AI*, though, does not have to generate the whole artifact at once

## QUIZ ME

Make a practice test. Do not show the answers yet

## GIVE ME A HINT

Help with the next step without solving the task

## ARGUE AGAINST ME

Build the strongest objection to my position

## LET ME TEACH YOU

Pretend you do not know; ask me questions

## CHECK MY WORK

Find the first unsupported or incorrect step

## PLAY YES-AND-NO WITH ME

Build the strongest case for both sides before synthesizing

## FACT-CHECK

Separate the claims; verify sources, entailment, and uncertainty

## BRAINSTORM WITH ME

Generate options without choosing the objective for me

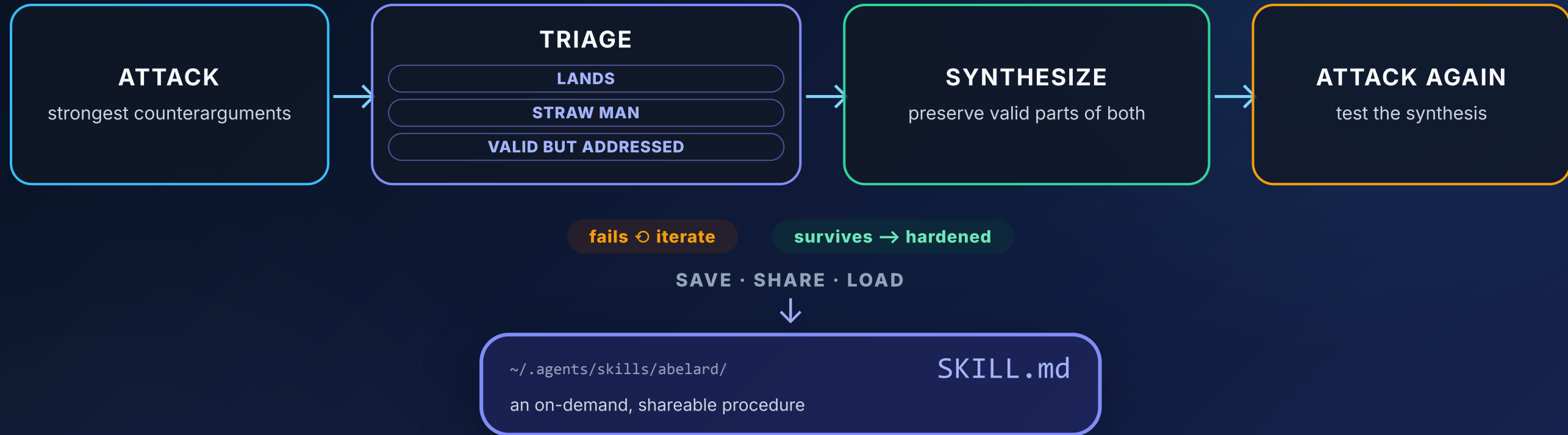
## WHAT SHOULD I KNOW?

Tell me something I probably do not know but should

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**ASK × 9 → GENERATE × 1** · Generate responsibly, or add more low-quality slop to the noosphere

# "Play yes-and-no" can be an actual SKILL



A SKILL makes one alternative as easy to invoke as the default; the other registers can become SKILLs too

# Our *epistemic honesty* protocol

## **LABEL THE STATUS**

Known, inferred, or unknown; keep uncertainty visible

## **CHECK THE CLAIM**

Test premises; verify sources, wording, entailment, and date

## **SEARCH AGAINST IT**

Find the strongest contrary case

## **NAME THE WITNESS**

What observation would change the answer?

## **EXTERNAL VERIFICATION**

Not validation: another mind, reproduction, or experiment

## **EXCUSE YOURSELF EASILY**

Acknowledge · analyze · log · move on

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This is *my* protocol for software development and AI research. Yours may be different

# What do you actually want?

Get JUST the answer and go to sleep?

Convince Frau Teacher or Herr Boss?

Test AI capability or calibration?

Get validation or support?

Understand the principle for later?

Explore philosophy with no immediate action?

Have some fun?

Humiliate the AI to feel better?

We are asking the AI to help with **WHAT?**

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If you do not choose the objective, AI will choose **for you** a plausible default, based on model priors and your register

# The more abstract the question, the more responsibility stays with you

$$x^2 - 3x + 2 = 0$$

$$x = ?$$

delegate

**Install Linux**

specify + verify

**Write an essay**

own purpose + authorship

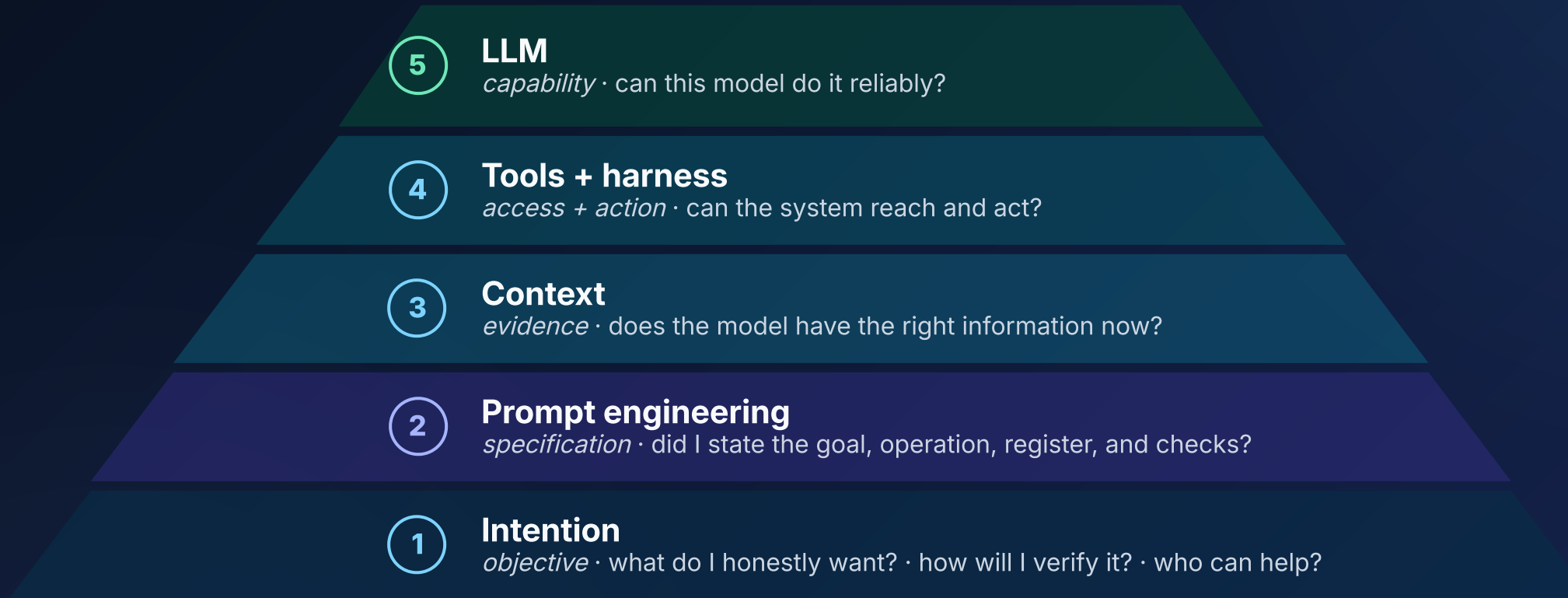
**What should I ask?**

own the choice

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Discuss the border with AI. The decision remains yours

# Which layer limits the result?



**AI FEEDBACK**  
probe capability



improve tools /  
harness



adjust context



refine specification



clarify intention

The model generates the prose; the whole stack determines whether it is useful

# Procedures you can inspect and edit

## EPISTEMIC HONESTY · INSTRUCTIONS

Label the status; check the claim; search against it; name the witness; seek external verification; correct openly

## ABELARD · SKILL

Attack; triage each objection; synthesize the valid parts; attack again



[eodus.github.io/thinking-with-ai/](https://eodus.github.io/thinking-with-ai/)

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Use them, change them, reject them. They are procedures — not universal truth

# AI can help with almost any question

You remain responsible for **why you asked** — and **what you do with the answer**

*Generate responsibly*

## So, questions?